

KEENE DILLANT-HOPKINS, NH, USA

WMO: 726165

Lat: 42.900N

Lon: 72.267W

Elev: 147

StdP: 99.58

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94721

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.0	(c) -17.5	(d) -28.3	(e) 0.3	(f) -18.6	(g) -24.1	(h) 0.4	(i) -15.6	(j) 9.0	(k) -3.9	(l) 8.2	(m) -5.3	(n) 0.9	(o) 190	(p) 0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.3	(c) 31.4	(d) 21.9	(e) 29.8	(f) 21.0	(g) 28.0	(h) 20.0	(i) 23.5	(j) 28.8	(k) 22.6	(l) 27.5	(m) 21.6	(n) 26.2	(o) 3.5	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.1	26.4	21.1	16.1	25.3	20.0	15.0	24.3	70.7	28.7	67.0	27.1	63.5	26.2	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.2	6.0	-24.5	33.8	3.5	1.5	-27.1	34.9	-29.1	35.7	-31.1	36.6	-33.6	37.6		
			-24.6	24.9	3.3	1.2	-27.0	25.7	-28.9	26.4	-30.7	27.1	-33.1	27.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-5.6	-3.9	0.7	7.1	13.3	18.3	21.1	20.1	15.9	9.4	3.4	-2.0		
	DBStd	10.29	6.31	5.71	5.19	4.50	4.12	3.59	2.95	3.06	3.99	4.40	4.83	5.16		
	HDD10.0	1933	483	391	293	110	13	0	0	0	2	66	204	373		
	HDD18.3	3932	741	624	547	337	165	44	8	17	94	277	447	631		
	CDD10.0	1283	0	0	4	23	114	250	346	312	179	47	7	1		
	CDD18.3	243	0	0	0	1	9	44	96	71	22	1	0	0		
	CDH23.3	2400	0	0	6	25	171	465	898	626	197	12	0	0		
	CDH26.7	710	0	0	1	6	48	137	296	169	51	1	0	0		
(14) Wind	WSAvg	1.9	2.1	2.2	2.4	2.4	2.1	1.8	1.6	1.4	1.5	1.8	1.9	1.9		
(15) Precipitation	PrecAvg	1004	74	61	76	83	96	88	93	94	89	89	89	81		
	PrecMax	1503	223	170	140	173	277	208	190	229	230	420	156	225		
	PrecMin	680	17	2	17	20	27	13	26	8	7	21	32	18		
	PrecStd	186	44	28	30	34	49	47	41	43	49	66	32	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.8	22.1	27.2	31.0	32.7	33.7	32.6	31.2	25.0	19.0	15.1		
		MCWB	11.7	8.7	13.3	15.2	19.0	22.3	23.9	22.8	21.3	17.7	15.0	12.3		
	2%	DB	7.7	9.0	15.2	22.5	27.7	30.2	31.7	30.3	27.7	22.2	17.1	10.1		
		MCWB	5.1	5.2	9.4	13.1	17.6	20.8	22.7	21.8	20.0	16.7	13.4	7.6		
	5%	DB	4.1	6.3	11.4	18.9	24.9	27.9	29.9	28.6	25.8	19.6	14.0	7.5		
		MCWB	1.8	2.9	6.4	11.8	16.2	19.7	21.5	21.0	18.7	15.8	10.5	5.2		
	10%	DB	2.5	3.7	8.0	16.3	22.3	26.2	27.9	27.2	23.0	17.3	11.9	5.0		
		MCWB	0.7	0.9	4.4	9.2	14.5	18.1	20.4	19.8	17.5	13.3	8.7	2.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	10.4	14.2	17.7	21.0	24.0	25.3	24.5	22.9	20.1	16.6	12.4		
		MCDB	12.8	12.3	19.2	22.7	27.7	30.0	31.6	29.7	28.3	22.4	18.2	13.8		
	2%	WB	5.2	5.7	10.2	15.0	19.1	22.5	23.8	23.1	21.5	18.0	14.1	8.1		
		MCDB	7.5	8.5	13.9	20.6	24.9	27.8	29.4	27.3	25.8	20.5	16.8	9.3		
	5%	WB	2.7	3.1	7.1	12.5	17.6	21.3	22.8	22.1	20.2	15.8	11.5	5.5		
		MCDB	4.1	5.9	10.5	17.3	23.0	25.8	27.8	26.9	23.9	18.9	13.4	7.4		
	10%	WB	0.9	1.4	4.8	10.4	15.8	20.0	21.7	21.1	18.9	13.8	8.7	3.1		
		MCDB	2.2	3.3	8.1	15.5	20.5	24.0	26.3	25.3	22.4	16.9	11.2	4.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.8	10.9	12.6	12.9	12.3	12.3	12.4	12.7	11.9	10.1	9.1		
		MCDBR	10.4	13.2	16.0	18.6	18.2	15.3	14.5	14.1	14.9	15.3	13.5	11.3		
	5% WB	MCWBR	8.8	9.9	10.4	10.3	9.1	7.7	6.8	6.7	8.4	10.0	10.3	9.5		
		MCWBR	9.9	12.5	14.3	15.9	15.2	12.7	12.6	12.0	12.4	13.0	11.7	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.293	0.321	0.362	0.395	0.435	0.456	0.425	0.375	0.334	0.305	0.283		
		taud	2.423	2.398	2.427	2.380	2.328	2.236	2.187	2.260	2.382	2.512	2.545	2.515		
	Ebn at Noon		859	912	925	905	879	841	817	832	853	855	834	835		
		Edh at Noon	75	91	101	115	125	137	143	128	105	82	68	64		
(44) All-Sky Solar Radiation	RadAvg		1.70	2.56	3.65	4.57	5.07	5.47	5.74	5.17	4.20	2.66	1.83	1.35		
	RadStd		0.13	0.22	0.24	0.51	0.59	0.52	0.37	0.27	0.30	0.27	0.17	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	+38

Nomenclature: See separate page